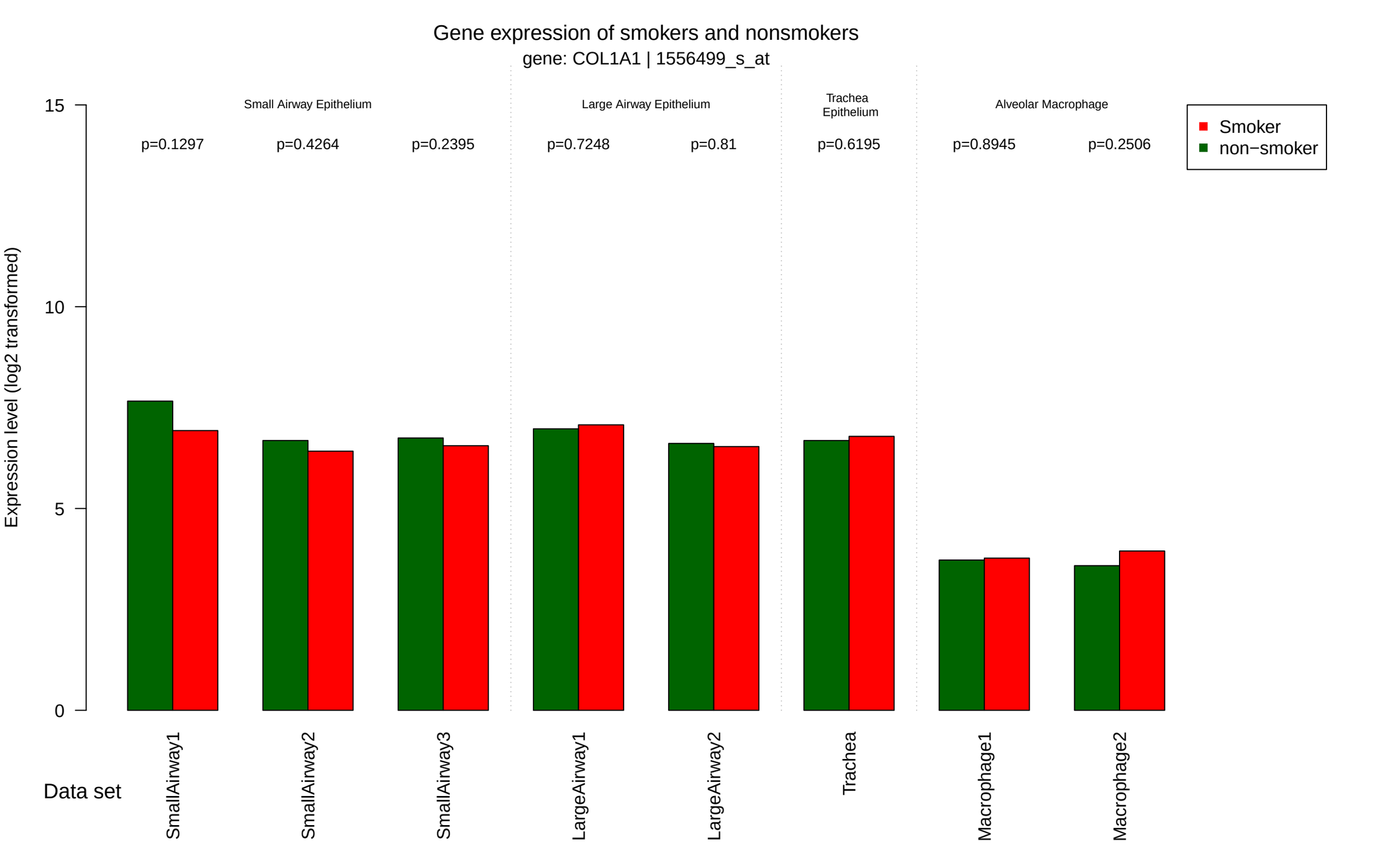
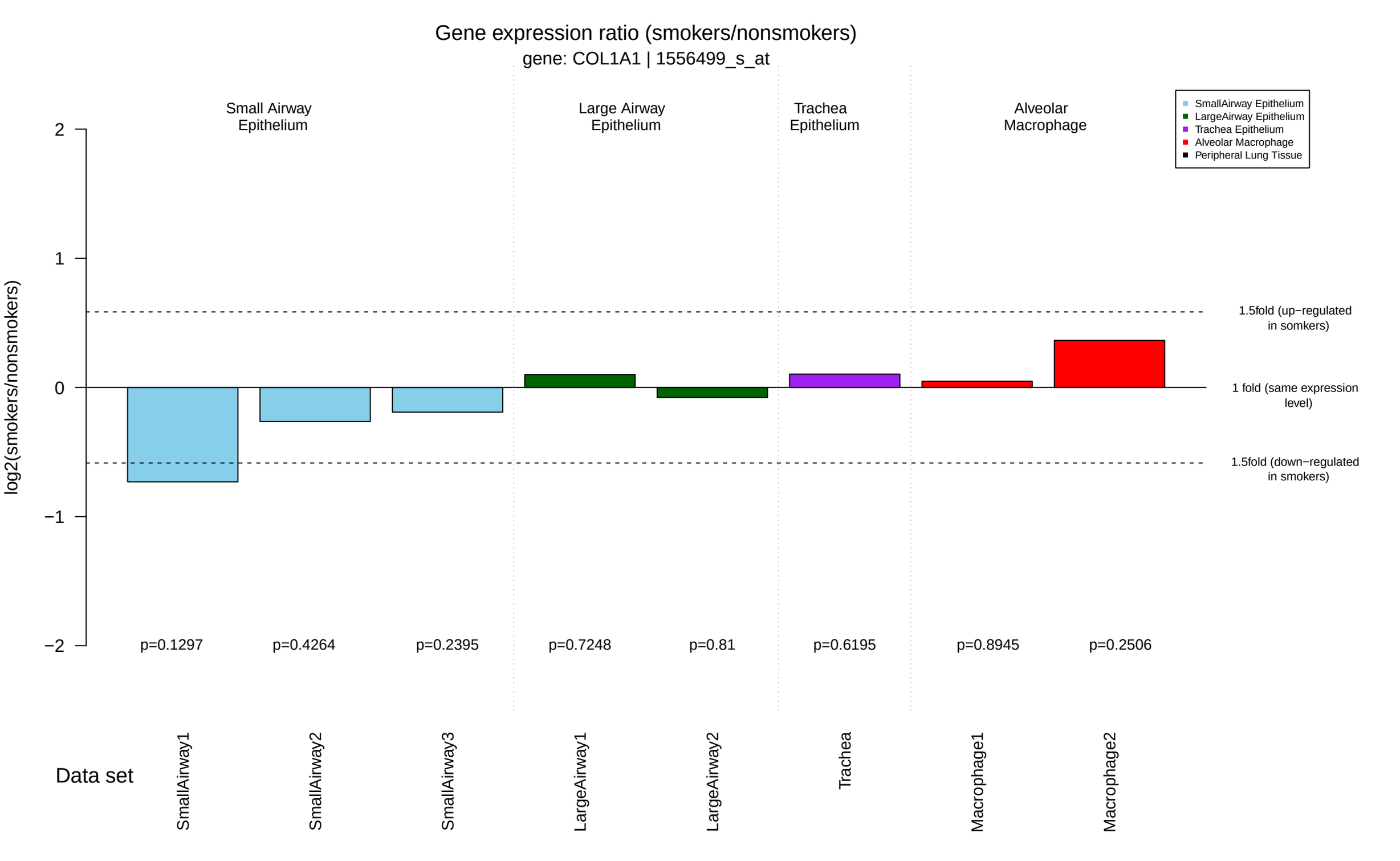
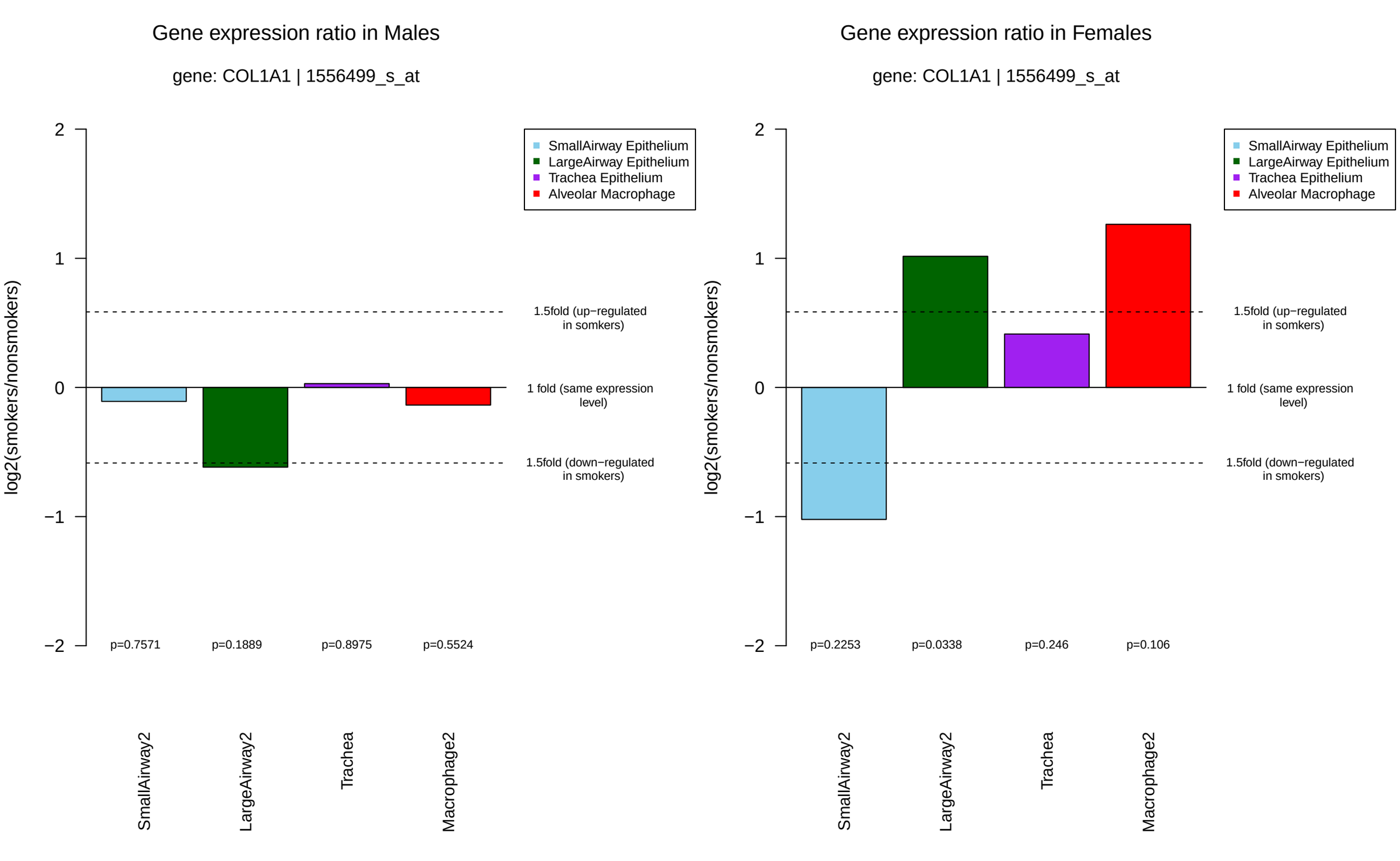
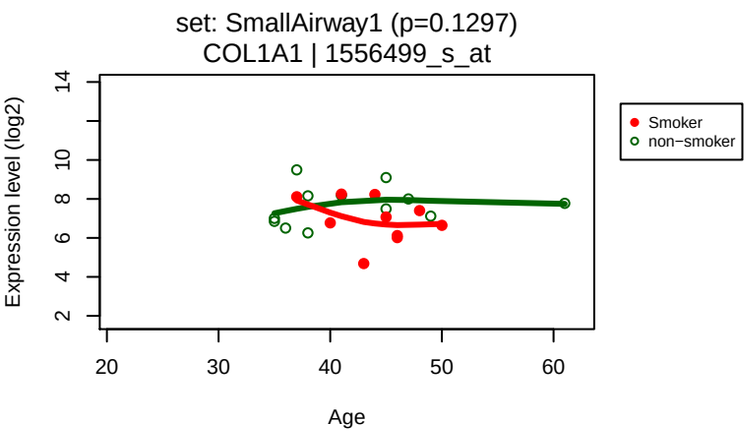
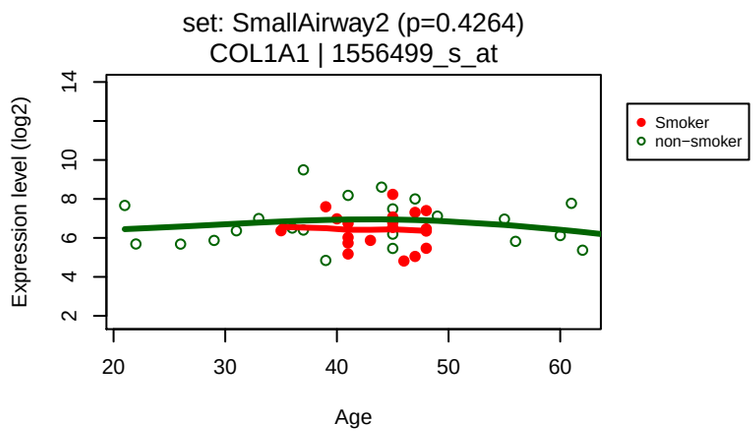
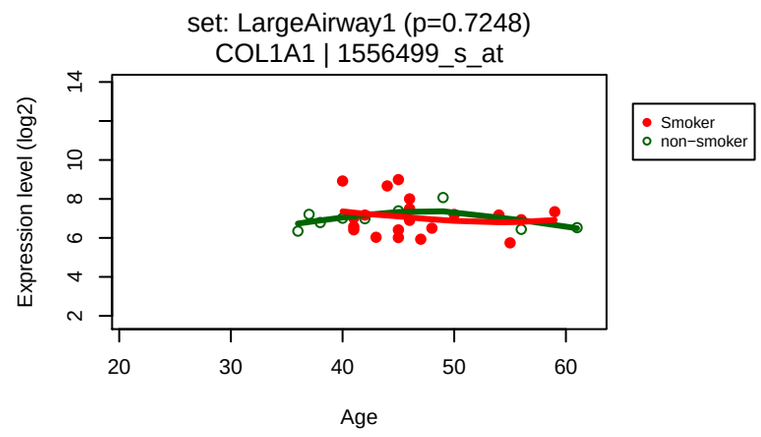

Gene expression of smokers and nonsmokers
gene: COL1A1 | 1556499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1297
p=0.4264
p=0.2395
p=0.7248
p=0.81
p=0.6195
p=0.8945
p=0.2506
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

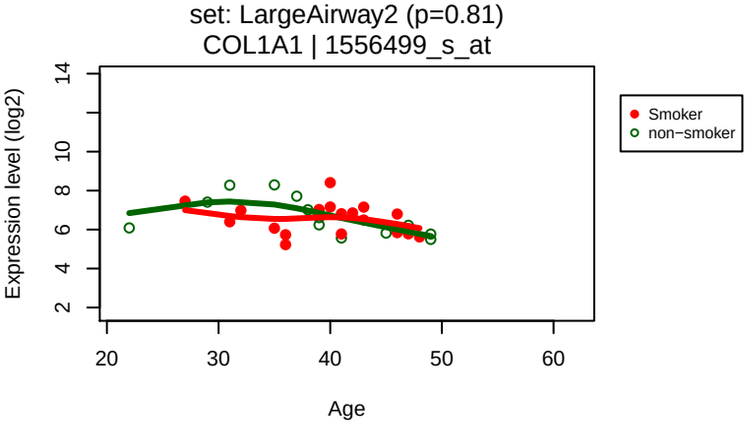

Gene expression ratio (smokers/nonsmokers)
gene: COL1A1 | 1556499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1297
p=0.4264
p=0.2395
p=0.7248
p=0.81
p=0.6195
p=0.8945
p=0.2506
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

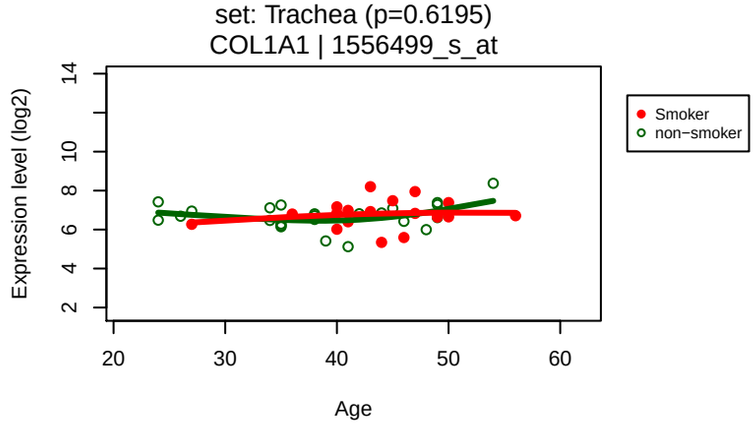

Gene expression ratio in Males
gene: COL1A1 | 1556499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7571
p=0.1889
p=0.8975
p=0.5524
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A1 | 1556499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2253
p=0.0338
p=0.246
p=0.106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

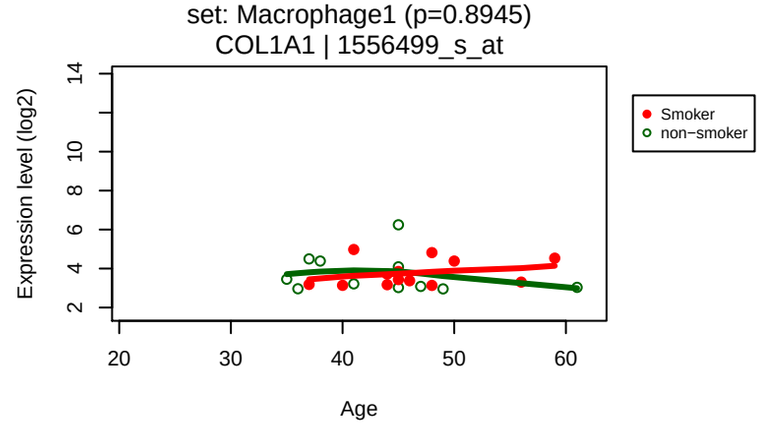

set: SmallAirway1 (p=0.1297)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker

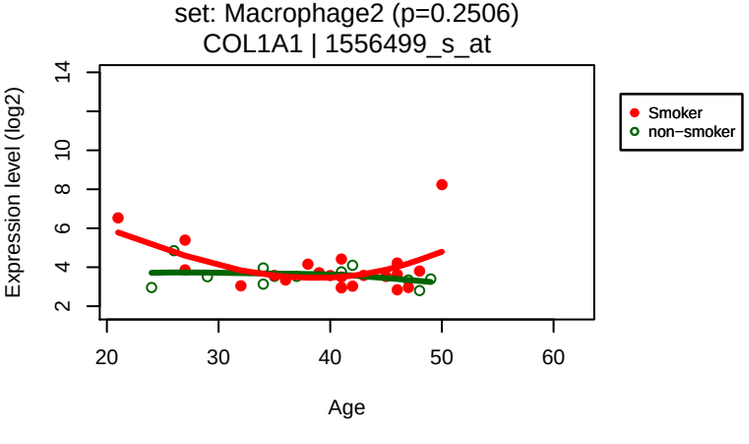

set: SmallAirway2 (p=0.4264)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker

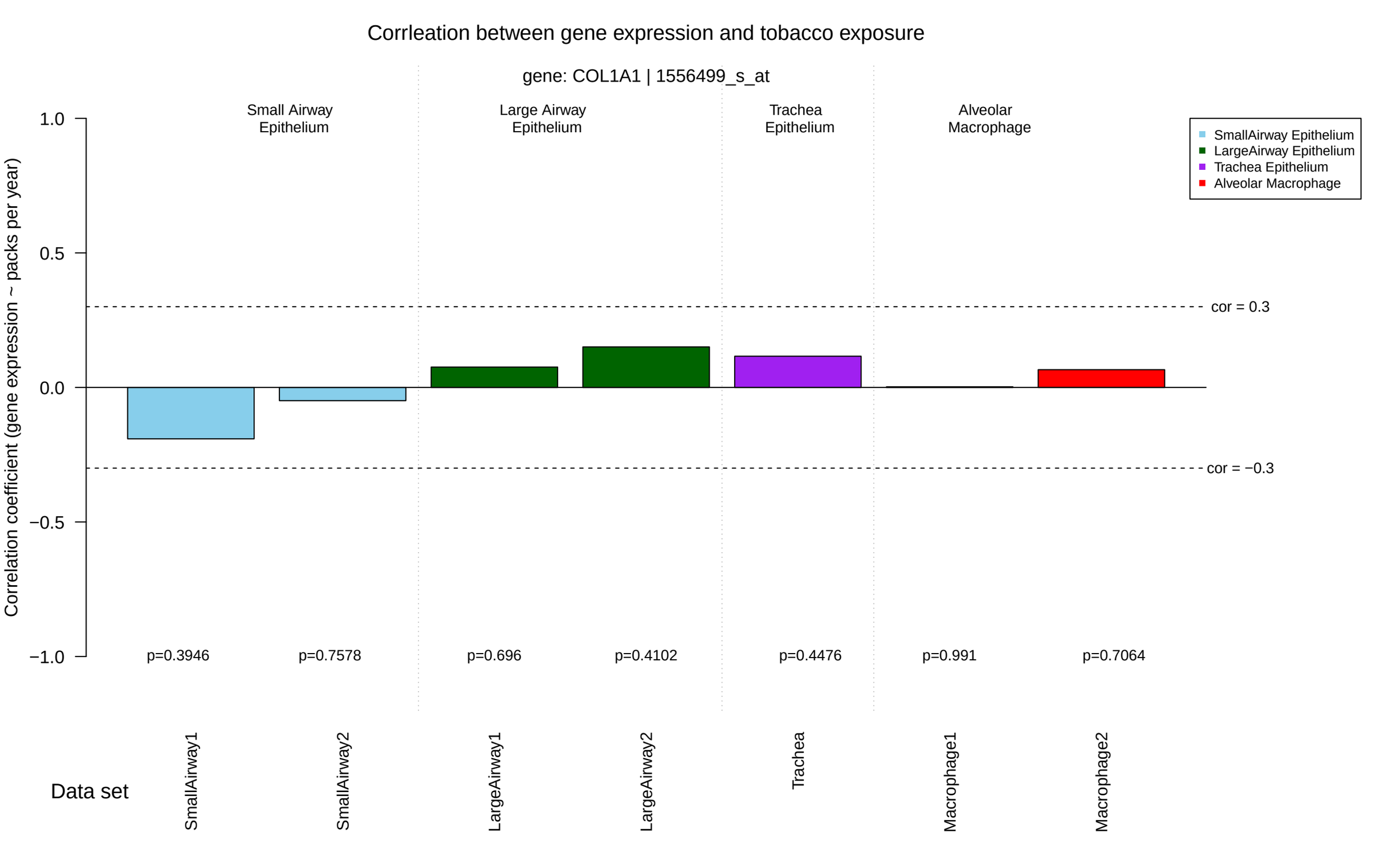

set: LargeAirway1 (p=0.7248)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.81)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6195)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8945)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2506)
COL1A1 | 1556499_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A1 | 1556499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3946
p=0.7578
p=0.696
p=0.4102
p=0.4476
p=0.991
p=0.7064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2